1 In lines 23 and 24 it says that "if the Pythagorean distance between two points A and B is d(A,B) then the taxicab distance satisfies the inequalities $\mathrm{d}(\mathrm{A},\mathrm{B}) \leqslant \mathrm{t}(\mathrm{A},\mathrm{B}) \leqslant \sqrt{2} \times \mathrm{d}(\mathrm{A},\mathrm{B})$."

This question is about using this result in generalised taxicab geometry.

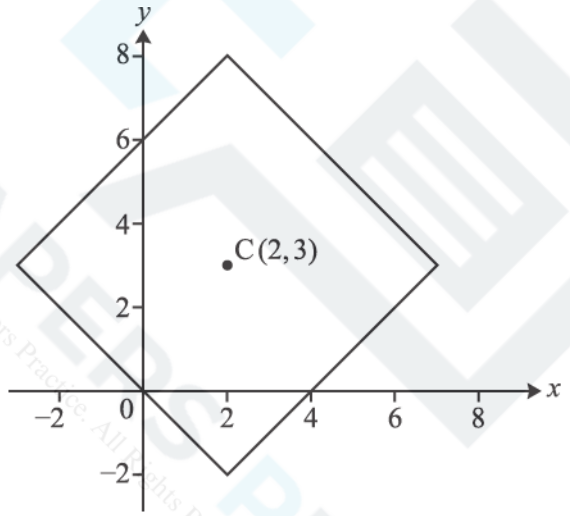
(i) Given that A is the point (0,0), describe all possible positions of B for which d(A,B) = t(A,B).

[1]

(ii) Given that A is the point (0,0), describe all possible positions of B for which $\mathrm{t}(\mathrm{A},\mathrm{B}) = \sqrt{2} \times \mathrm{d}(\mathrm{A},\mathrm{B})$.

[2]

2 Fig. 7 is reproduced below.

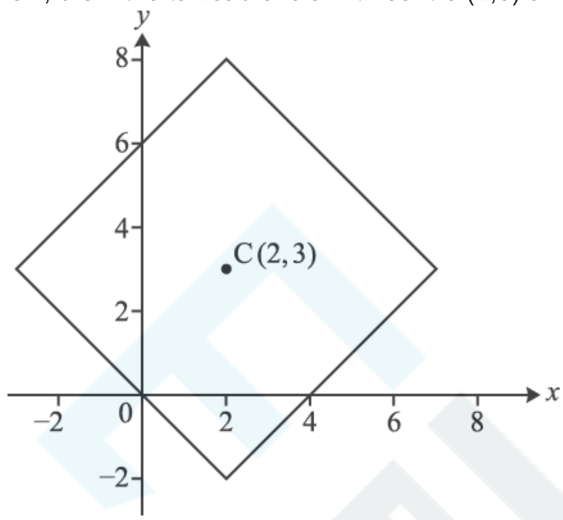


(i) Two points on this locus have *x*-coordinate – 0.7. Write down the coordinates of each of these points.

[2]

(ii) In lines 77 to 78 it says “adding a second taxicab circle with centre (2,0) and radius 2 shows that in generalised taxicab geometry two different circles can have an infinite number of points in common!”

On the copy of Fig. 7 given below, draw the taxicab circle with centre (2,0) and radius 2.



[1]

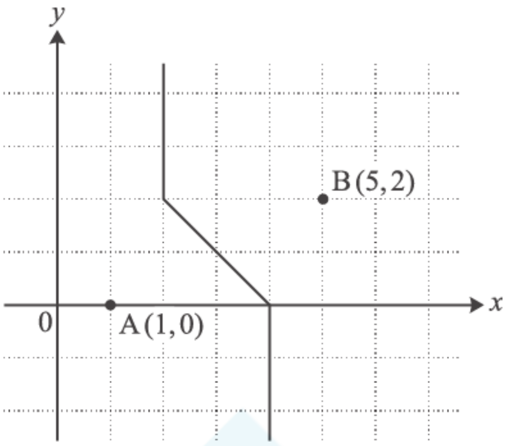
3 In lines 54 and 55 it says there are 35 minimum distance routes from A (0,0) to B (4,3). Determine how many of these routes pass through the point with coordinates (3,2), explaining your reasoning.

[2]

4 Referring to Fig. 5, or otherwise, find the value of n(4,4).

[2]

(i) Describe the following locus of a point P, using the notation t(P,A) and t(P,B) as appropriate.



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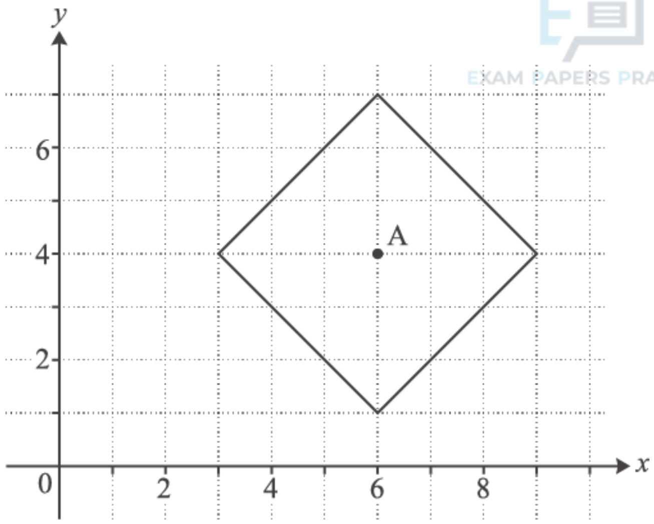
[1]

(ii) Describe the following locus of a point P, using the notation t(P,A) as appropriate.

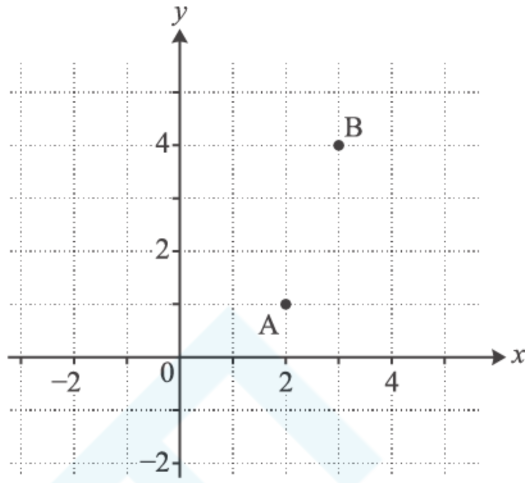


[2]

6 This question is concerned with generalised taxicab geometry.

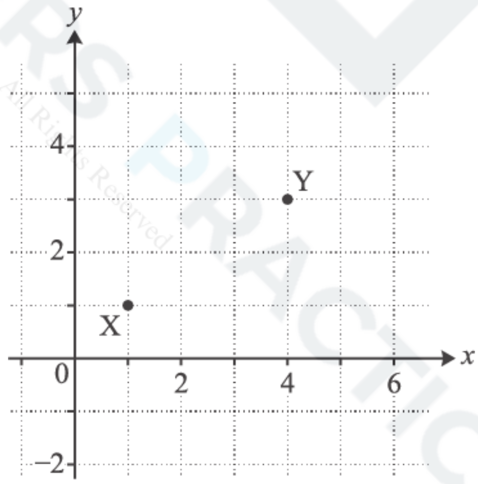
On the grid below, show the locus of a point P where t(P,A) = t(P,B).

[3]

2	

7 On the grid below mark all three possible positions of the point P with integer coordinates for which t(P,X) = 4 and t(P,Y) = 3.

[3]

1	

(i) Use Equation (3) to calculate the declination of the sun on February 2nd.

[3]

(ii) The town of Boston, in Lincolnshire, has latitude 53° north and longitude 0°.

Calculate the time of sunset in Boston on February 2nd.

Give your answer in hours and minutes using the 24-hour clock.

[4]

9 In lines 94 and 95 the article says

"Fig. 8 shows you that at latitude 60° north the terminator passes approximately through time +9 hours and ?9 hours so that there are about 18 hours of daylight."

Use Equation (4) to check the accuracy of the figure of 18 hours.

[4]

10 The graph is a copy of Fig. 6.

The article says that it shows the terminator in the cases where the sun has declination 10° north, 1° north,5° south and 15° south.

Identify which curve (A, B, C or D) relates to which declination.

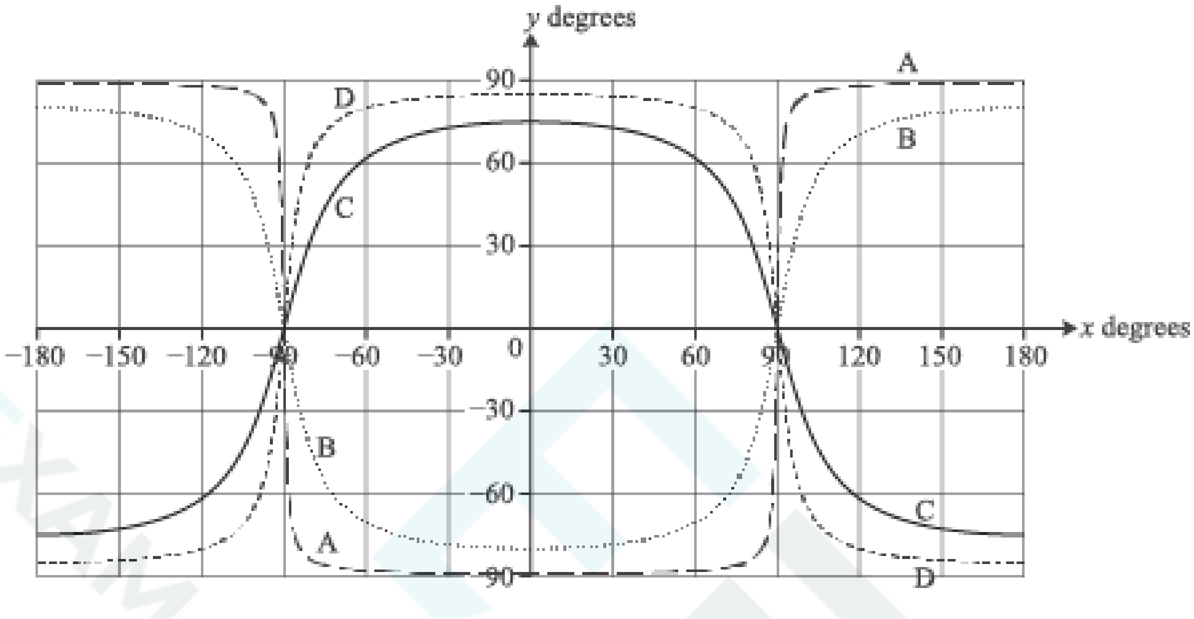


Fig. 6

10° north:

1° north:

5° south:

15° south:

[2]

11 Use Fig. 8 to estimate the difference in the length of daylight between places with latitudes of 30° south and 60° south on the day for which the graph applies.

[3]

12 The diagram is a copy of Fig. 4.

R is a place with latitude 45° north and longitude 60° west. Show the position of R on the diagram.

M is the sub-solar point. It is on the Greenwich meridian and the declination of the sun is $+20°$. Show the position of M on the diagram.

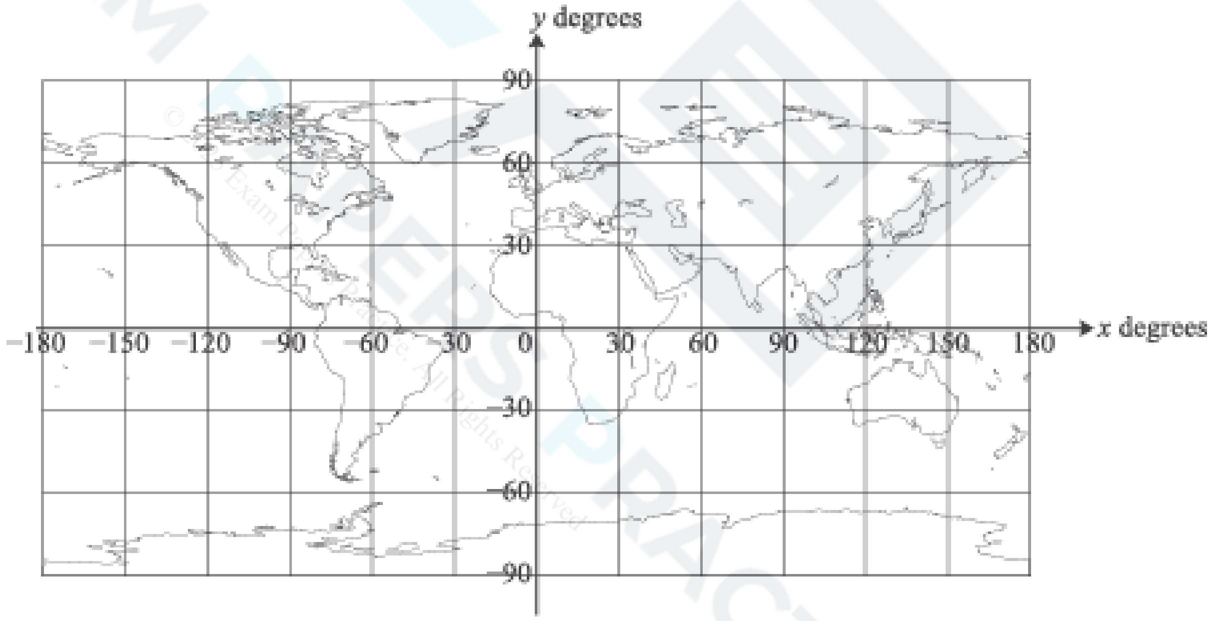


Fig. 4

[2]

13 In line 96, the article says

‘As a result of the study, it was recommended that a focal length of 75 mm should be used.’

Make a reasoned estimate of the percentages of participants in Stirling University's study who would have thought the photomontages made the wind turbines appear ‘Too large’, ‘About right’ and ‘Too small’ if a lens of focal length 75 mm had been used. You must state your assumptions clearly.

[4]

14 In the diagram, the wind turbine BT is observed from two different positions P and Q. The blade tip height of the turbine is 72 m.

Both P and Q are a horizontal distance of 800 m from the turbine.

P is at the same height as the base, B, of the turbine. Q is 18 m below the level of B.

The angle of elevation from P is α; the angle TQB is β.

Show that the angles α and β, in degrees, are the same to 2 significant figures.

[3]

15 The diagram illustrates the situation in Fig. 5 of the article.

The blade tip height of the wind turbine is 99.5 m.

The base, B, of the turbine is 120 m higher than A and at a horizontal distance of 320 m from A.

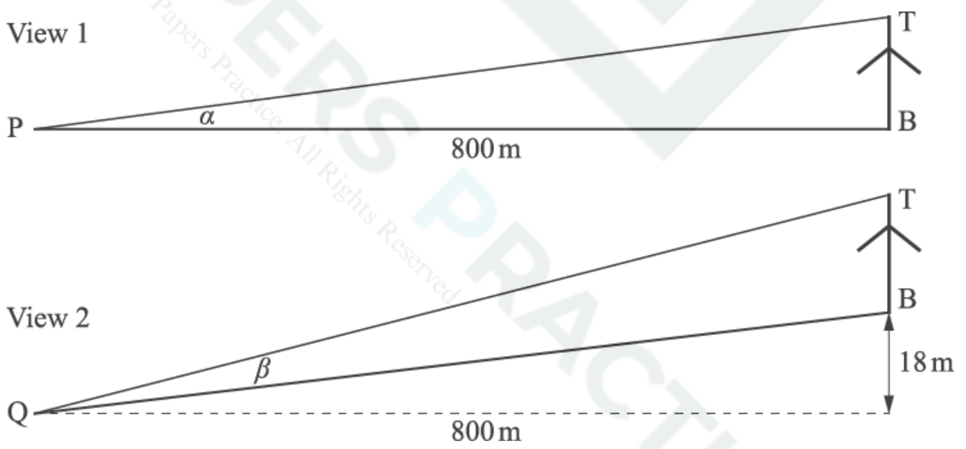
An observer at A can see the top 20 m of a blade when it is pointing vertically upwards, as in the diagram.

The observer's line of vision is a tangent to the hill at C. The horizontal distance from A to C is 140 m.

Find the height of C above A.

[4]

Not to scale



16 A wind turbine with a blade tip height of 125 m is seen from a distance of 623 m. The ground is level and horizontal so that the whole of the turbine can be seen.

(i) Calculate the angle of elevation of the tip of a blade when it is pointing vertically upwards. You should assume that the viewer's eye is at the same height as the base of the turbine.

[1]

The wind turbine is shown on a photomontage; the viewing distance is stated to be 51.4 cm.

(ii) Calculate the height that the turbine would have on the photomontage if it were seen with the same angle of elevation as that in part **(i)**.

[1]

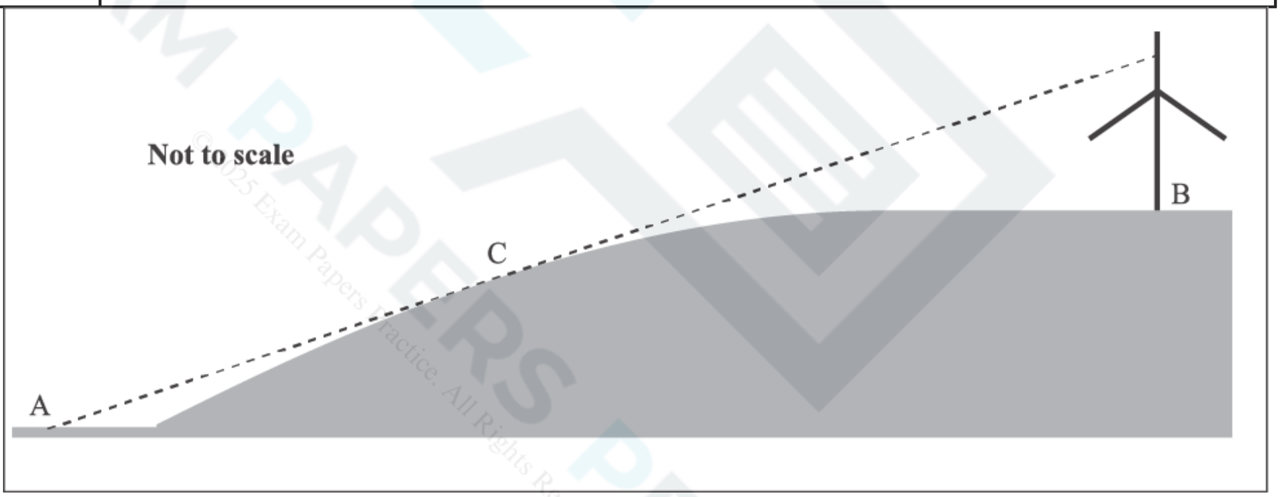
The image of the wind turbine is 7.3 cm high when the photomontage is printed on A4 paper.

(iii) Show that when the photomontage is printed on A3 paper, the height of the wind turbine is consistent with the angle of elevation found in part **(i)**.

[2]

(i)	
(ii)	

(iii)



17 In lines 46 and 47, the article says

‘So someone at the point of observation would not see the bottom 12 m of the turbine.’

Explain how the figure of 12 m was obtained.

[2]

18 The blades of a wind turbine sweep out a circle of diameter 90 m. The turbine's blade tip height is 149.5 m.

Calculate the hub height of this turbine.

[1]

19 (See Insert for Practice 64003.) Use integration by parts to show that $\int_1^8 \ln x \, dx = 8 \ln 8 - 7$, as given in line 13. **[4]**

20 (See Insert for Practice 64003.) Starting from Fig. C1.3, show that $\ln 8! > 8 \ln 8 - 7$, as given in line 19. **[3]**

21 **(a)** (See Insert for Practice 64003.) Prove the following.

(i) The function $\ln x$ is increasing throughout its domain. **[2]**

(ii) The graph of $y = \ln x$ is concave downwards. **[2]**

(b) (See Insert for Practice 64003.) Explain why the approximation to $\int_1^n \ln x \, dx$ using the trapezium rule must be an underestimate of the exact value of the integral. **[1]**

22 (See Insert for Practice 64003.) ${}^{2n}C_n = \dfrac{(2n)!}{n!n!}$ □□ is sometimes called the nth central binomial coefficient.

Using Stirling's formula as given in line 37, show that ${}^{2n}C_n \approx \dfrac{4^n}{\sqrt{\pi n}}$. **[3]**

23 (See insert for June 2017 475401B) State the set of values of x_0 for which the iteration

$$x_{n+1} = 2.5x_n(1 - x_n)$$

(i) converges to a single non-zero number, **[1]**

(ii) has all terms from x_1 onwards equal to zero. **[1]**

24 (See insert for June 2017 475401B)

(i) Use the algebraic method indicated in lines 68 to 70 to find the equilibrium point of the iteration

$$x_{n+1} = 1.6x_n (1 - x_n).$$

[2]

(ii) Show that the iteration

$$x_{n+1} = x_n^2 + 2$$

does not have any points of equilibrium. **[2]**

25 (See insert for June 2017 475401B) One of the assumptions for the model used for the population of squirrels in the text was that there are no predators.

An alternative model is proposed in which predators kill a fixed number of squirrels each year.

An iterative equation for this model is given by

$$x_{n+1} = kx_n (1 - x_n) - 0.25.$$

In the table below x_0 is taken to be 0.55 and four different values are considered for k.

(i) Complete as many of the empty cells as you need to in order to establish the outcomes for these values of k.

	$x_{n+1} = kx_n(1-x_n) - 0.25$			
	$k = 2$	$k = 3$	$k = 4$	$k = 5$
x_0	0.55	0.55	0.55	0.55
x_1	0.245	0.4925	0.74	0.9875
x_2				
x_3				
x_4				
x_5				
x_6				
x_7				
x_8				
x_9				
x_{10}				
…	…	…	…	…

(ii) Comment on what the table tells you for each of the four values of *k*. **[6]**

26 (See insert for June 2017 475401B)

(i) Table 3 gives the first four points of bifurcation of the iteration

$$x_{n+1} = kx_n(1 - x_n).$$

Feigenbaum's Constant is 4.6692 correct to 5 significant figures. Using this value for the ratio of the interval lengths, estimate the values of k for the next two points of bifurcation. **[3]**

(ii) (A) Find, S, the sum to infinity of the geometric series

$$1 + \frac{1}{4.6692} + \left(\frac{1}{4.6692}\right)^2 + \left(\frac{1}{4.6692}\right)^3 + \dots.$$

[2]

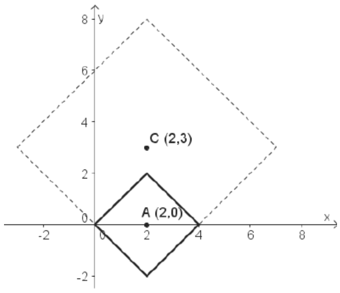
(B) Using certain figures from Table 3, a value of k is estimated to be

$$k = 3.5644 + 0.0203 \times S.$$

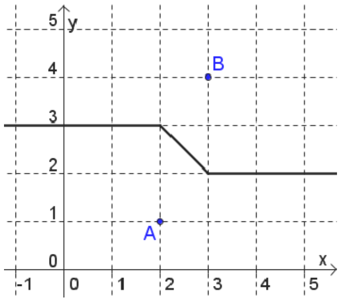
State what happens at this value of k. **[1]**

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Part marks and guidance	
1		i	B must lie on one of the two axes	B1	oe, say, horizontal and vertical lines from (0,0).	Do not accept a list of points unless an overall statement that includes **all** points is given (not just integers).
		ii	B lies on line $y = x$	B1	oe	As **(i)**
		ii	or on line $y = -x$	B1	**Examiner's Comments** There were many correct solutions using mathematical terms such as the x-axis, y-axis, x=0, y=0, y=x and y= $-x$. Some others gave appropriate descriptions such as all points (p,0) for all real values p or said all points vertically above and below and horizontally from A or equivalent whilst others used the points of the compass in their descriptions. Some only gave a list of points which was insufficient without a general statement to include all points. Some only gave integer values. In several cases one felt that the candidates probably did know the correct answer but were unable to explain it clearly.	
			Total	3		
2		i	(–0.7,5.3)	B1	**Examiner's Comments** There were some wrong answers with unclear methods although there were also many correct solutions.	SC B1 for (y =) 5.3 and 0.7.
		i	(–0.7,0.7)	B1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		B1	A square with straight edges as shown with a vertex at (2,2). (and nothing more)	
		ii			**Examiner's Comments** This part was ususally right.	
			Total	3		
3			The 10 routes shown from (0,0) to (3,2) each continue in one way via (4,2) to (4,3) and each continue in one way via (3,3) to (4,3)	2		Or, the 35 all pass through either 20(3,3) or 15 (4,2). 10 of the 20 at (3,3) come from (3,2) [and the rest from (2,3)] 10 of the 15 at (4,2) come from (3,2) [and the rest from (4,1)] So 10 + 10 = 20 are from (3,2) oe
			Hence 20	B2	www Need explanation SC B1 without explanation **Examiner's Comments** This was less successful. Many gave the answer 10 as *n*(3,2) is 10. Others gave 10+2=12.	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
4			Evidence of 35 + n(3,4) 70	M1 A1	70 www gets B2. **Examiner's Comments** Many obtained both marks although some left their answer as n(3,4)+n(4,3) without evaluation.	
			Total	2		
5		i	t (P,A) = t (P,B)	B1	(not t(P,A) = t(P,B) = 3) **Examiner's Comments** This was usually correct although some stated t(P,A)=t(P,B)=3 which was wrong.	Accept words, such as, [distance] t(P,A) is equal to [distance] t(P,B).
		ii	t (P,A) = 3	B1	**Examiner's Comments** t(P,A)=3 was usually given.	Accept words such as, all the points with [distance] t(P,A) equal to 3.
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
6				B3	B3: Exactly as shown (with line extended beyond (4,2) and before (0,3)) Award B1 for a locus including one partially correct line segment or at least two correct discrete points in that segment Award B2 for a locus that contains parts of all three correct line segments (or at least two discrete points within each of them) and no additional incorrect points. **Examiner's Comments** This was generally well answered although some just gave one point. Some got the sloping part right but had vertical, not horizontal, lines. Others had the right horizontals but the sloping line was flatter.	
			Total	3		
7			(4,0) and (5,1) and (2,4)	B3	All three points correct (and no additional points) Accept a list of coordinates. **Examiner's Comments** Most candidates scored all three marks.	SC B1 any correct point SC B2 any two correct points (and no more than two incorrect additional points) B0 if points unclear
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
8		i	$\alpha = -23.44 \times \cos\left(\frac{360}{365} \times (n+10)\right)$			
		i	On February 2nd, $n = 31 + 2 = 33$	B1	calculate $n = 31 + 2 = 33$ (days in Jan + Feb) **soi**	SC B1 condone $30 + 2$
		i	$\alpha = -23.44 \times \cos\left(\frac{360}{365} \times 43\right)$	M1	substitution of their $n + 10$ in equation (3) and attempt to evaluate	Where $n = 31, 32, 33, 34$ only
		i	$a = -17.31$	A1	or -17.306 or rounds to -17.3 **Examiner's Comments** Most candidates found ? = -17.31 as required. A few chose to use the number of days in January as 30 and lost one mark. Those who thought February was the first or third month received no credit.	NB $n = 32 + 10$ gives -17.576 gaining B1M1A0
		ii	$\tan y = -\frac{1}{\tan \alpha}\cos(15t)$			
		ii	$\tan 53 = -\frac{1}{\tan(-17.306)} \times \cos(15t)$	M1	use of **their** α in equation (4)	
		ii	$t = \frac{1}{15}\arccos(-\tan 53 \times \tan(-17.306))$	DM1	making t the subject	
		ii	$t = 4.3717$	A1	cao	**SC ft** from -17.576 Obtains A1ft for 4.343 (or 4.34)
		ii	Sunset is at 12 hrs + 4 hours 22 minutes, and so 16:22 hrs	A1		And then A1ft for 16:21 (or 16:20)

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii			<u>Examiner's Comments</u> Some candidates carelessly lost the negative sign in their angle or the negative sign in the formula and so lost unnecessary marks. Many correctly obtained *t* =4.37 but not all converted this correctly to the 24 hour clock. 16:37 was commonly seen. Candidates were able to follow through for full marks from the Special Case in part(ii)	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
9			$\tan y = -\dfrac{1}{\tan\alpha}\cos(15t)$			
			? = 23.44°, y = 60°, t is to be found			
			$\cos(15t) = -0.7509\ldots$ $15t = 138.6737\ \ldots$	M1	substitute in formula and attempt to solve (as far as $15t$ = invcos … …) oe	
			$t = 9.2449\ \ldots$	A1	accept 9.2 or better	
			Daylight hours are 2 × 9.2449… = 18.4898 …	DM1	doubling, dependent on first M1	
			So 18.5 hours (to 3s.f.)	A1	or approx 18 hours www (accept 18.4898, 18.489, 18.49, 18.5 or 18.4 (from 2 × 9.2), 18.48)	any reasonable accuracy or stating error is approx 0.49 oe
			OR			
			Using t = 9 ? = 23.44, t = 9, y is to be found	M1	t = 18/2 = 9 and substituted in formula	any reasonable accuracy
			$\tan y = 1.6309$……..	DM1	and attempt to solve (as far as y = inv tan constant) oe	
			y = 58.485° so approx 60°	A2	or 58.49° / 58.5° / approx 60° www **<u>Examiner's Comments</u>** Most candidates substituted α =60° and found t=9.2449 or similar and then the majority multiplied it by 2 to compare it with18. A few worked backwards from t=9 to reach approximately 60° when substituting in the appropriate equation, and were given full credit.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
10			10° north is B 1° north is A 5° south is D 15° south is C	B2	All four answers correct **SC B1** Any two answers correct Examiner's Comments The graphs were usually identified correctly although there were also many guesses. The response tended to be either fully correct or all wrong.	
			Total	2		
11			For 30° S, 10 hours daylight	B1	cao soi	10 only
			For 60° S, 5.5 hours daylight.	B1	allow $5 < t < 6$ soi	not 5 or 6
			Difference 4.5 hours	B1ft	10-t dependent on both previous B marks soi Examiner's Comments Most candidates had the right idea but some were inaccurate with 10, 6, 4 being the most common alternative solution.	**SC(1)** allow B3 for $4 <$ difference in length < 5 (**without wrong working**) **SC(2)** allow B2 if uses $t = 5$ or $t = 6$ eg, 10,6,4 www **SC(3)** If calculates (using the equation (4)) can obtain all three marks. Approximate values are 10.07, 5.51, 4.56. (May also work with 5 and $2.5 < t < 3$ if doubled at end, e.g. 5 B1 dependent on later doubling 2.6 B1 dependent on later doubling $(5 – 2.6) \times 2 = 4.8$ B1)
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
12			Point R marked correctly at (–60, 45) Point M marked correctly at (0, 20)	B1 B1	half way up the relevant square Generously applied if above half way If unclear, B0B0. __Examiner's Comments__ Answers were often correct but surprisingly many, and various, incorrect positions were seen. A number of candidates only used the letter *R* or *M* to indicate their points where the addition of a cross or dot would have made their position clearer.	Need labelling with letters or co-ordinates oe (condone one not labelled if other is labelled (and no others marked)) **SC B1** BOD both marked (and no others) and no labelling.
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
13			Those thinking they appear **about right** might be those who opted for **70** mm and **80** mm, so 85 + 85 = **170** people	B1	Assumption stated explicitly and must mention the bolded values / words (for 170 allow 85 + 85 or use of 85 twice – may be seen in % calculation) – **any reasonable assumptions must be justified** (as in the case above) – it must be clear where the numbers are coming from (e.g. 220 is about right which comes from 60, 70 and 80 would score B1)	
			Percentage $=\frac{170}{362}\times 100 = 46.96...$ so 47%	B1	For 47% (or better) **or** 23% (or better – accept 23.5%) (comes from using just the value of 85) **or** a percentage which comes from a justified assumption (so if the % is not 47 or 23 then they must have been awarded the first B mark)	
			Those saying **too large** would be people who opted for **50** mm or **60** mm, so 16 + 50 = **66** people Those saying **too small** would be people who opted for **90** mm, **100** mm or **110** mm, so 72 + 37 +17 = **126** people	B1	Assumptions stated explicitly and must mention the bolded values / words (allow 'values less than or equal to 60' for 50 and 60, allow 'values greater than or equal to 90' for 90,100 and 110) – see first B mark for further details. **Note that for this mark the 'too large' and 'too small' must be attributed to the correct values**	
			Percentages are 18% (too large) and 35% (too small)	B1	For 18% (or better) and 35% (or better) **or** a percentage which comes from a justified assumption (so if the % are not 18 and 35 then they must have been awarded the third B mark). **Note that for this mark the values of 18 and 35 do not need to be attributed correctly but any**	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					other values must correspond correctly to 'large' and 'small'	

Focal length	50	60	70	80	90	100	110
Number of …	16	50	85	85	72	37	17
%	4.419…	13.81…	23.48…	23.48…	19.88…	10.22…	4.696…

Examiner's Comments

While many candidates made reasonable estimates of the percentages of participants many confused the two cases of 'too large' and 'too small', or they did not make the numbers used in their calculations clear. A number of candidates did not calculate any percentages but instead gave the total number of participants in the three different groups. The most common estimates were that those who opted for 70 mm and 80 mm would believe that 75 mm was 'about right', while those who opted for 50 mm and 60 mm would say 'too large' and finally those who opted for 90 mm, 100 mm and 110 mm would say 'too small'. Finally many candidates did not state any of their assumptions clearly even though this was specifically asked for in the question.

			Total	4	

Question			Answer/Indicative content	Marks	Part marks and guidance	
14			$\alpha = \arctan\left(\frac{72}{800}\right) \quad (=5.142...)$	B1	Beware: QB = 800.2	
			$\beta = \arctan\left(\frac{90}{800}\right) - \arctan\left(\frac{18}{800}\right) \quad (=5.129...)$	B1	Or cosine rule e.g. $\cos\beta = \frac{\left(\sqrt{800^2+18^2}\right)^2 + \left(\sqrt{800^2+90^2}\right)^2 - 72^2}{2\left(\sqrt{800^2+90^2}\right)\left(\sqrt{800^2+18^2}\right)}$ (oe)	
			The angles are both 5.1° to 2 significant figures.	E1	**Must see 5.14 (or better), 5.13 (or better) and 5.1** **Examiner's Comments** Nearly all candidates correctly stated that $\alpha = \arctan\left(\frac{72}{800}\right)$ although many assumed incorrectly that triangle TQB was right-angled. Of those that did realise that triangle TQB was not right-angled many took the slightly more long-winded approach of using the cosine rule to find β instead of realising that Finally, many candidates did not realise that if they are required to show that two answers agree to 2 significant figures then they must quote both value correct to 3 significant figures and so in this case examiners needed to see as a minimum of α = 5.14 and β = 5.13 followed by 5.1.	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
15			Height above A of lowest visible point is	M1	All three values 99.5, 120, 20 seen in a calculation – 199.5 with no working scores M1A1	
			99.5 + 120 – 20 = 199.5 m	A1		
			Using similar triangles or trigonometry	M1	$\frac{x}{320}\times 140$ **or** $\tan\theta = \frac{x}{320}$ and 140 tan θ where179.5 ≤ x ≤ 219.5 (must be a complete method for finding the height)	
			Height of C $= \frac{199.5}{320}\times 140 = 87.3\text{ m}$	A1	87.3 (or better) **Examiner's Comments** The responses to this question were mixed with many candidates failing to find the height above A of the lowest visible point as 199.5 m with many incorrectly using a height of 219.5 m. Furthermore, many candidates failed to read the question carefully and stated that the distance AC was 140 m rather the horizontal distance from A to C.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
16		i	$\tan\alpha = \frac{125}{623} \Rightarrow \alpha = 11.3°$	B1	11.3 or better - allow 11.35 rounded to 11.4 but B0 if only 11.4 seen **Examiner's Comments** Parts (i) and (ii) were nearly always correct although in part (iii) many candidates failed to use the value of 7.3 to show that when the photomontage was printed on A3 paper, the height of the wind turbine was consistent with the angle of elevation found in part (ii).	
		ii	$\text{Height } = \frac{125}{623} \times 51.4 = 10.3 \text{ cm}$	B1	10.3 or better	
		iii	1.41 × 7.3 or their(10.3) ÷ 1.41	M1	1.41 or better $\left(\text{e.g. } \sqrt{2}, \frac{297}{210}\right)$	
		iii	1.41×7.3 =10.3 or 10.3÷1.41= 7.3 or $\tan^{-1}\left(\frac{7.3\times'1.41'}{51.4}\right) = 11.3...°$	A1	Must see 10.3 (or better) or 7.3 (or better) – 10.3 with no working scores M0A0 11.3 or 11.4	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
17			Scale Photomontage: Real size = 2.35 cm : 35 m 0.8 cm corresponds to $\frac{0.8 \times 35}{2.35} = 11.91...$ rounding to 12 m	M1 E1	values 2.35, 35 and 0.8 **or** 6.7, 99.5 and 0.8 **or** 99.5, 35, 2.35 and 5.9 seen in a calculation or $\frac{0.8 \times 99.5}{6.7} = 11.88...$ **- in both cases must see 11.9 (or better) followed by** **12** or $99.5 - \frac{35}{2.35}(5.9) = 11.627...$ leading to 12 so **must see 11.6 (or better) followed by 12** (or equivalent correct calculation) **Examiner's Comments** While the majority of candidates correctly explained how the figure of 12 m was obtained with the most common method being via the calculation $\frac{0.8 \times 99.5}{6.7}$ which lead to a value of 11.88… many did not give sufficient detail or tried to justify the figure of 12 without showing any calculation at all.	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
18			Blade radius = 45 m ⇒ Hub height = 149.5 – 45 = 104.5 m	B1	**Examiner's Comments** Nearly all candidates correctly stated the hub height of the turbine although a number incorrectly found this height as 59.5 m (which came from using the diameter of the blade rather than the radius).	
			Total	1		
19			$\int_1^8 \ln x \, dx = [x \ln x]_1^8 - \int_1^8 \frac{x}{x} \, dx$	M1(AO 1.1a)	Use of integration by parts with 1 and ln x	
				A1(AO 1.1)	May not include limits at this stage	
			$= [x \ln x - x]_1^8$	M1(AO 2.5)		
			= 8ln8 – 8 – (1ln1–1) = 8ln8 –7 **AG**	E1(AO 2.1)	All correct including limits	
				[4]	Convincing completion	
			Total	4		
20			Area of rectangles = ln 2 + ln 3 +…+ ln8	B1(AO 1.1)		
			= ln(2×3×…×8)	M1(AO 2.2a)		
			Area of rectangles $> \int_1^8 \ln x$ dx,	E1(AO 2.1)	Convincing completion	
			so ln 8! > 8ln8 –7 **AG**	[3]		
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance		
21		a	$(A)\quad f'(x)=\frac{1}{x}$ oe $x > 0$ [throughout domain] so gradient is positive, i.e. ln x is increasing	M1(AO 1.2) E1(AO 2.1) [2]	Correct derivative Convincing completion of argument		
		b	$(B)\quad \frac{d^2y}{dx^2}=-\frac{1}{x^2}$ oe This is negative so curve is concave downwards	M1(AO 1.2) E1(AO 2.1) [2]	Correct derivative Convincing completion of argument		
		c	Since the curve is concave downwards, for each trapezium there is part of the area under the curve which is not included in the trapezium	E1(AO 2.4) [1]	Or other convincing explanation ; e.g. The shape of the curve means that all the trapeziums are below it	The explanation must refer to *all* the trapeziums	
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance	
22			$\frac{(2n)!}{n!n!} \approx \frac{\sqrt{4\pi n}\left(\frac{2n}{\mathrm{e}}\right)^{2n}}{\left(\sqrt{2\pi n}\right)^2\left(\frac{n}{\mathrm{e}}\right)^{2n}}$	M1(AO 1.1)	Use of Stirling's formula	
			$= \frac{2\sqrt{\pi n}}{2\pi n}\left(\frac{2n\mathrm{e}}{n\mathrm{e}}\right)^{2n}$	M1(AO 3.1a)	Correct simplification of either the power terms or the 'π*n*' terms	$\frac{1}{\sqrt{\pi n}} \times \frac{\left(\frac{2n}{\mathrm{e}}\right)^{2n}}{\left(\frac{n}{\mathrm{e}}\right)^{2n}}$
			$= \frac{2^{2n}}{\sqrt{\pi n}} = \frac{4^n}{\sqrt{\pi n}}$ **AG**	A1(AO 2.1) [3]	Correct completion	
			Total	3		
23		i	(All x_0 for which) $0 < x_0 < 1$	B1 [1]	Condone as separate inequalities – allow x, x_n, etc. for x_0 Accept (0, 1) (i.e. correct set-notation) – must be strict inequalities	
		ii	$x_0 = 0$ or 1	B1 [1]	Both required – allow x, x_n, etc. for x_0 - condone 0, 1 stated without x_0 (0 and 1 must not appear as part of a range of values) **Examiner's Comments** Both parts of this first question on paper B proved to be surprisingly demanding for many candidates with only a minority correctly stating that in part (i) $0 < x_0 < 1$ and that in part (ii) $x_0 = 0$ or $x_0 = 1$.	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
24		i	$x = 1.6x(1 - x)$ $x(1.6x - 0.6) = 0$ $(x = 0 \text{ or})\ x = \frac{0.6}{1.6} = 0.375 \text{ or } \frac{3}{8}$	M1 A1 [2]	For correctly replacing x_{n+1} and x_n with x - must be using 1.6 Using an iterative approach or stating the correct answer with no working - M0 A0 Accept as a minimum the correct equation stated in terms of x only followed by the correct answer for both marks **Examiner's Comments** Even though in part (i) the question specifically asked for candidates to find the equilibrium point by using the algebraic method indicated in the text, many candidates incorrectly used an iterative method. For those candidates that stated $x = 1.6x(1 - x)$ most went on to solve this equation correctly.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$x^2 - x + 2 = 0$ Discriminant = $(-1)^2 - 4\times1\times2 = -7$ or $x = \dfrac{1 \pm \sqrt{(-1)^2 - 4(1)(2)}}{2}$ so no (real) roots (oe)	M1 A1 [2]	Re-arranging **and** correct attempt at either the discriminant or solving using the correct formula/completing the square for their 3-term quadratic equation. Allow consideration of $\sqrt{b^2 - 4ac}$ with correct substitution of their values only Correct working with a correct conclusion (so if discriminant stated must be –7 but may be awarded for $1 - 4(2) \Rightarrow$ no (real) roots / no points of equilibrium) No marks for non-algebraic approach **Examiner's Comments** In part (ii) whilst many candidates correctly stated that at the point of equilibrium $x = x^2 + 2$ many did not give a mathematically sound reason for why this equation does not have any real solutions. It was insufficient to use an iterative approach and it was simply incorrect to state that the quadratic equation does not factorise. However, many candidates did re-arrange the equation correctly and then proceeded to show that the discriminant of the resulting three-term quadratic was negative.	
			Total	4		

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th colspan="2">Part marks and guidance</th></tr>
<tr><td>25</td><td></td><td>i</td><td>
<table>
<tr><th></th><th colspan="4">$x_{n+1} = kx_n(1-x_n) - 0.25$</th></tr>
<tr><th></th><th>$k = 2$</th><th>$k = 3$</th><th>$k = 4$</th><th>$k = 5$</th></tr>
<tr><td>x_0</td><td>0.55</td><td>0.55</td><td>0.55</td><td>0.55</td></tr>
<tr><td>x_1</td><td>0.245</td><td>0.4925</td><td>0.74</td><td>0.9875</td></tr>
<tr><td>x_2</td><td>0.1199...</td><td>0.4998...</td><td>0.5196</td><td>−0.1882...</td></tr>
<tr><td>x_3</td><td>−0.0388...</td><td>0.4999...</td><td>0.7484...</td><td>...</td></tr>
<tr><td>x_4</td><td>...</td><td>0.5</td><td>0.5030...</td><td>...</td></tr>
<tr><td>x_5</td><td>...</td><td>...</td><td>0.7499...</td><td>...</td></tr>
<tr><td>x_6</td><td>...</td><td>...</td><td>0.5000...</td><td>...</td></tr>
<tr><td>x_7</td><td>...</td><td>...</td><td>0.7499...</td><td>...</td></tr>
<tr><td>x_8</td><td>...</td><td>...</td><td>0.5000...</td><td>...</td></tr>
<tr><td>x_9</td><td></td><td></td><td>0.75</td><td></td></tr>
<tr><td>x_{10}</td><td></td><td></td><td>0.5</td><td></td></tr>
<tr><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td></tr>
</table>
</td><td>B1
B1
B1
B1</td><td>There may be some variation in the columns for $k = 3$ and $k = 4$. Some calculators will round to the final values earlier than others. In general answers must be given to at least 2dp

Allow correct truncated or rounded answers

One mark for each column taken far enough

$k = 2$ correct to at least x_3
$k = 3$ correct to at least x_3 (accept 0.5)
$k = 4$ correct to at least x_6 (accept 0.5)
$k = 5$ correct to at least x_2

isw after reaching the values for each column as stated above

Examiner's Comments

This question was answered extremely well with many candidates scoring all four marks in part (i). The vast majority of candidates scored the three marks for filling in the table correctly for k = 2,3 and 5 but for the case when k = 4 a number of candidates did not complete a sufficient number of empty cells to clearly establish the outcome for this value.</td><td></td></tr>
</table>

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	For k = 2 it is **unbounded** or '**dying out**' For k = 3 **it converges** (to x = 0.5) or **stable** or **equilibrium** For k = 4 it **oscillates** (between x = 0.5 and x = 0.75) or **alternates** but not **fluctuates** For k = 5, it is **unbounded** or '**dying out**'	B2 [6]	All 4 correct Allow B1 for 2 correct Allow equivalent to 'dying out' provided unambiguous Must be one of these three words Must be one of these two words Allow equivalent to 'dying out' provided unambiguous **Examiner's Comments** Most candidates in part (ii) commented extremely well on what the values in the table implied about the four values of k; the main error being in the use of incorrect phrases or terminology (e.g. using 'fluctuating' to describe the behaviour when k = 4).	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
26		i	$3.5644+\frac{0.0203}{4.6692}$	B1	Correct method – allow a slip in one value (e.g. 4.692) only – either correct value for *k* implies this mark	
			= 3.5687	B1	Accept 3.569	
			$3.5687+0.0203\times\left(\frac{1}{4.6692}\right)^2=3.5696$ to 5 sf	B1	(3.5687476…) – no ft from incorrect values – accept if given as the upper bound of an interval	
				[3]	Accept 3.5697 or 3.570 but not 3.57 – no ft from incorrect values – accept if given as an upper bound **Examiner's Comments** Candidates seemed to be evenly split in part (i) in either making no real progress when attempting to find the values of *k* (for the next two points of bifurcation) or they correctly found both. Where issues occurred it was mainly due to accuracy.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	(A) $S = \dfrac{1}{1 - \dfrac{1}{4.6692}}$ $S = 1.2725\ldots$	M1 A1 [2]	Use of the correct formula for the sum to infinity of a GP – allow a slip in the value of r only (e.g. 4.692) – allow r stated to 2dp or better ($r = 0.21416945\ldots$) Accept 1.273 not 1.27 (1.2725389…) M0 A0 if formula not used **Examiner's Comments** In part (ii)(A) it was expected that candidates would use the sum to infinity formula for a geometric series and not use the less mathematically sound approach of simply adding up the first few terms on their calculator. Furthermore, it was surprising in this part how many candidates either struggled with finding the common ratio (many claiming that $r = 4.6692$ rather than the correct $r = \dfrac{1}{4.6692}$) or gave a final answer to a degree of accuracy which was not appropriate to the given context.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	(*B*) It is an estimate of a value of k for which chaos is occurring	B1 [1]	Accept any mention of 'chaos' **Examiner's Comments** Finally, in part (ii)(*B*), while many candidates left this part blank or simply worked out the value of k it was encouraging that many candidates correctly stated that at this value chaos would be occurring.	
			Total	6		